

4. Built an Autonomous "Audible" Loop

Overview

A real quarterback doesn't just run the play called in the huddle if the defense shows an unbeatable look. We gave our AI the same capability. The Flask backend now evaluates the expected EPA of its primary play call against the inputted defensive coverage. If the math projects failure, it autonomously vetoes the call and audibles to a better option.

Code Changes

- **Backend (Flask):** Added dynamic threshold checks in app.py. After the `pbp_situation_model` determines the primary play type, the server runs an EPA simulation.
- **Logic:** if `expected_epa < 0.0`: was implemented to trigger an override, flipping the play type (e.g., from Run to Pass) and recalculating the visual and mathematical outputs before sending the final payload back to React.

Future Roadmap

We plan to expand the complexity of the audibles. Instead of just flipping from a generic Run to a generic Pass, we will program the AI to "check" into specific concepts (e.g., if a Run is called against an 8-man box, the audible will specifically force a Quick Screen or Slant route rather than a deep dropback pass).